Algebra 2 Review: Exponential & Logarithmic Functions

Answer Key

Station 1: Exponential Functions: Evaluating, Graphing, and Transformations

1. $\frac{1}{2}$
2. 22
3. Decay; 15% decay per period
   since $b = 1 - r$ with $0 < b < 1$
4. Growth; 20% growth per period
   since $b = 1 + r$ with $b > 1$
5. $y = 6(1.5)^x$
6. $y = 5 \cdot 3^x$
   $a = 5$; $b^2 = 9$
7. $y = 2 \cdot 3^x$
8. Domain: all reals; range: $y > 5$; asymptote $y = 5$
9. Left 2, reflect over the x-axis, up 1; asymptote $y = 1$
10. y-intercept $(0, 2)$; asymptote $y = -2$

Station 2: Compound Interest, the Number e, and Continuous Growth

1. \$4,266.30
   $3000(1.045)^8$
2. \$9,096.98
3. \$2,023.92
4. \$3,597.69
   $2500e^{0.052(7)}$
5. 2.7169; about 0.0014 below e
   larger n gets closer to e
6. 6% quarterly, by \$14.91
   \$5,387.42 vs. \$5,372.51
7. 13.0 years
8. 9.9 years
9. $k \approx 0.231$; about 5,040 bacteria
10. 78.5%

Station 3: Logarithms: Definition, Evaluating, Properties, and Change of Base

1. $\log_2 32 = 5$
2. $9^2 = 81$
3. -5
4. $\frac{2}{3}$
   $8^y = 4 \Rightarrow 2^{3y} = 2^2$
5. 2
   $4 + (-2)$
6. $2 + 2\log_3 x - \log_3 y$
7. $2 + 3\log a + \frac{1}{2}\log b$
8. $\log_5 12$
9. 2.312
10. $\log 20 \approx 1.301$ but $\log 10 + \log 10 = 2$
    the product rule gives $\log(ab)$, not $\log(a + b)$

Station 4: Solving Exponential and Logarithmic Equations

1. $x = 2$
   $4^{x+1} = 4^3$
2. $x = 3$
   $2^{3x} = 2^{x+6}$
3. $x = 5$
   $2x - 1 = 3x - 6$
4. $x = \frac{1 + \ln 15}{2} \approx 1.85$
5. $x \approx 2.010$
6. $x = 5$
   $2x - 1 = 3^2$
7. $x = 3$
   both arguments equal 7
8. $x = 4$
   $x = -2$ is extraneous
9. $x = 1$
   $x = -10$ is extraneous
10. 21.8 years